SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA0302	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	Slot-A	**Date:**	

Code of Conduct

IJ.Raju(192325117) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: J.Raju

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.

A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.
2. Determine which query contains semantic interpretation errors.
3. Evaluate how meaning representation affects chatbot decision accuracy.

4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

2. First-Order Predicate Calculus for Smart Manufacturing.
An Industry 4.0 manufacturing plant monitors machines using semantic rules.
Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- Active(x) → Producing(x)
- Produces(x,y) ∧ Active(x) → Available(y)
- Maintenance(x) → ¬Producing(x) **Tasks:**

1. Represent the production data using First-Order Predicate Calculus.
2. Analyze which products are currently available.
3. Infer whether Gear production is affected by maintenance activities.
4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

3. Word Sense Disambiguation in E-Commerce Search Engines.
An e-commerce company observed the following search logs.

"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object

User Query	Possible Sense 1	Possible Sense 2

Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	**Role**
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							

4							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

1. Semantic Representation in Customer Support Chatbots

1. Action–Object Relationships

Customer Query	Representation	Action	Object
Activate international roaming for my number.	ACTIVATE(Roaming, Customer)	Activate	Roaming
Deactivate caller tune service.	DEACTIVATE(CallerTune, Customer)	Deactivate	Caller Tune
Check my data balance.	QUERY(DataBalance, Customer)	Query	Data Balance
Enable 5G service.	ACTIVATE(5GService, Customer)	Activate	5G Service

2. Query with Semantic Interpretation Error

Q2 contains the error. The actual intent is **Deactivate Caller Tune**, but the predicted intent is **Activate Caller Tune**. The correct representation is DEACTIVATE(CallerTune, Customer). This is a critical action error because activation and deactivation perform opposite operations.

3. Effect of Meaning Representation on Chatbot Accuracy

Meaning representation converts natural-language queries into structured information that the chatbot can process. Correct identification of the action, object, intent, and customer entity enables the system to perform the correct operation. An incorrect representation can lead to wrong responses or incorrect service operations. Therefore, accurate semantic parsing directly improves chatbot decision accuracy and customer satisfaction.

4. Improvements for Better Semantic Understanding

• Use context-aware NLP models such as transformer-based language models.

• Train with a large and diverse collection of telecom customer queries.

• Combine intent classification with named-entity recognition.

• Handle synonyms such as enable/activate and disable/deactivate.

• Use conversation history for follow-up queries.

• Add confidence scores and request confirmation when intent is uncertain.

• Continuously retrain using incorrectly classified queries.

• Use domain-specific telecom ontologies.

2. First-Order Predicate Calculus for Smart Manufacturing

1. Production Data Representation

Production facts: Active(M1), Active(M2), Maintenance(M3), Active(M4).

Rules: $\forall x\ \text{Active}(x) \rightarrow \text{Producing}(x)$; $\forall x \forall y\ (\text{Produces}(x,y) \land \text{Active}(x) \rightarrow \text{Available}(y))$; $\forall x\ \text{Maintenance}(x) \rightarrow \neg\text{Producing}(x)$.

From the rules: Producing(M1), Producing(M2), Producing(M4), and ¬Producing(M3). Thus, M1, M2, and M4 are producing, while M3 is not producing because it is under maintenance.

2. Products Currently Available

The exact available products cannot be identified because no specific Produces(x,y) facts are provided. If Produces(M1,Gear) and Active(M1) are given, then Available(Gear) can be inferred.

3. Is Gear Production Affected by Maintenance?

M3 is under maintenance, so ¬Producing(M3) follows. If M3 is responsible for producing Gear, then Gear production is affected. However, because Produces(M3,Gear) is not explicitly provided, the effect on Gear cannot be concluded with certainty.

4. Effectiveness of Predicate Logic

First-Order Predicate Calculus is useful for representing machines, products, statuses, and relationships in a structured way. Rules can automatically derive production states and support monitoring, planning, fault detection, and decision-making. Its main limitation is that conclusions depend on the accuracy and completeness of the facts and rules.

3. Word Sense Disambiguation in E-Commerce Search Engines

1. Correct Word Sense

Query	Context / Clicked Result	Correct Sense
Apple accessories	iPhone Charger	Technology Brand
Mouse wireless	Bluetooth Mouse	Computer Device
Java tutorial	Coding Lessons	Programming Language
Python course	Software Development Training	Programming Language

2. Semantic Cues

• Apple + accessories suggests electronic accessories rather than fruit.

• Mouse + wireless strongly suggests a computer mouse.

• Java + tutorial is strongly associated with programming.

• Python + course and software-development results indicate the programming language.

• Clicked-result behavior provides strong evidence of user intent.

3. Impact of Incorrect Sense Selection

Incorrect WSD reduces search relevance and can produce wrong recommendations. For example, interpreting Apple as fruit could return food products instead of iPhone accessories. This can lower click-through rate, conversion rate, customer satisfaction, and increase search abandonment.

4. Industrial-Scale WSD Strategy

• Preprocess queries and identify ambiguous words.

• Analyze surrounding words and query context.

• Use transformer-based contextual embeddings.

• Combine semantic similarity with product-category information.

• Use search, click, purchase, and conversion history.

• Use domain-specific knowledge graphs.

• Rank candidate meanings using confidence scores.

• Continuously retrain using real customer interactions.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems

1. Contribution of Syntax to Semantic Roles

Syntactic structure helps determine who performed an action, what was affected, and who received it. In “Doctor prescribed medicine to patient,” Doctor is the Agent, Medicine is the Theme, and Patient is the Recipient. In “Patient reported severe headache,” Patient is the Experiencer and Headache is the Symptom. In “Nurse monitored patient continuously,” Nurse is the Agent and Patient is the Object/Theme.

2. Appropriateness of Assigned Roles

Entity	Given Role	Correct Role	Assessment

Doctor	Agent	Agent	Correct
Medicine	Instrument	Theme	Incorrect
Patient	Recipient	Recipient	Correct
Headache	Symptom	Symptom	Correct

Medicine is the thing being prescribed, so it is the Theme, not an Instrument. An example of an Instrument would be a stethoscope in “Doctor examined the patient with a stethoscope.”

3. Errors Caused by Incorrect Parsing

• Wrong identification of the doctor or patient.

• Incorrect medication information.

• Incorrect medical relationships.

• Wrong symptom identification.

• Errors in electronic health records.

• Incorrect clinical decision support and information extraction.

4. Methods to Improve Healthcare Semantic Analysis

• Use domain-specific medical parsers trained on clinical text.

• Use clinical language models such as BioBERT or ClinicalBERT.

• Combine syntactic parsing with semantic role labeling.

• Use medical named-entity recognition for drugs, diseases, symptoms, and procedures.

• Use medical ontologies and knowledge graphs.

• Analyze document-level context rather than isolated sentences.

• Use confidence scoring and uncertainty detection.

• Train and evaluate with expert-annotated medical datasets using precision, recall, and F1-score.

Overall Conclusion

Semantic representation, predicate logic, word sense disambiguation, and syntax-driven semantic analysis are important NLP techniques. Semantic representation converts language into actionable meaning; predicate calculus supports rule-based reasoning; WSD resolves ambiguity; and syntactic analysis helps identify semantic roles. Combining these techniques with modern contextual NLP models can improve accuracy, automation, and decision support in telecom, manufacturing, e-commerce, and healthcare applications.